Tan Xue Wen

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Education

National University of Singapore (NUS) *Aug 2021 – Aug 2025**

Ph.D in Financial Technology

- GPA: 5.00/5.00
- Awarded the Asian Institute of Digital Finance (AIDF) Scholarship
  - The highest-tier scholarship at the time, valued at SGD 6,000 per month
- Teaching Assistant for FT5001 (Fintech Innovations for Consumers) under Prof Keith B. Carter
- Research Interest: Explainable AI, Large Language Models, AI for Social Good, Computational Linguistics
- Proposed Thesis: Human-Centered AI for Informed Financial Decision-Making

B.Eng in Electrical Engineering *Aug 2015 – Aug 2019*

- GPA: 4.67/5.00 (Highest Distinction / First Class Honours), Dean's List
- Alcatel-Lucent Technologies Prize Winner
- Final Year Project Grade: A+, 1st Prize Winner for ECE FYP Presentation 2019
- Undergraduate Research Opportunity Program (UROP)
- Fall Semester Exchange at National Taiwan University of Science and Technology
- Norminated member of ECE Undergraduate Student Council and peer Tutor for EE2024 (Programming for Computer Interfaces) under the ECE Peer Tutoring Scheme

Publications

ScamGPT-J: Inside the Scammer's Mind, *Dec 2024*
A Generative AI-Based Approach Toward Combating Messaging Scams
- **Xue Wen Tan**, Kenneth See, Stanley Kok

- Paper accepted at the International Conference on Information Systems (ICIS) 2024 in Bangkok, Thailand
- Poster accepted at The 3rd Quantum Computing Workshop 2024 in Singapore (No Proceeding)

Explainable Risk Classification in Financial Reports *Dec 2023*
- **Xue Wen Tan**, Stanley Kok

- Paper accepted at Internation Conference on Information System (ICIS) 2023 in Hyderabad, India
- Paper accepted at AAAI Summer Symposium AI4FinTech 2023 in Singapore
  - Did not archive in proceeding, extended to ICIS 2023
- Presented at ETH-NUS FinsureTech Conference 2023 in Zurich, Switzerland

**The International Conference on Information Systems (ICIS) is recognized as a top-tier conference in information systems research, holding an A+ ranking under the NUS Conference Ranking*

Working Papers Under Submission

Holistic Multiagent Systems Framework for Agile Software Development Lifecycle
- Kenneth See, **Xue Wen Tan**

- Currently under review at The 24th International Conference on Autonomous Agents and Multiagent Systems (pending acceptance), under Blue Sky Ideas special track

Explainable Joint Entity and Relation Extraction with Slot Attention
- **Xue Wen Tan**, Stanley Kok

- Currently under review at ACL Rolling Review, targeting paper acceptance at EMNLP or ACL

Experience

Technology Executive *Singapore*
DBS Bank Limited, Institutional Banking Group Technology *Aug 2019 – Aug 2021*

- Hired under the highly selective DBS SEED Graduate Programme, which required candidates to pass a rigorous hiring process, including coding assessments, IQ/EQ evaluations, a hackathon, and a panel interview

Tech Lead & IT Project Manager *Jan 2020 – Aug 2021*

- Delivered end-to-end digital solutions for Corporate Banking applications (IDEAL), including Audit Confirmation, Digital Banking for HK SMEs, InvoiceNow for automated supplier payments, Trusple integration with ANT Group, Stop Payment, Foreign Exchange, Subsequent Account Opening, Contextual Marketing, Customer Self-Administration, and the migration of Indonesia applications to ensure regulatory compliance, *many of which have been featured in media outlets such as Business Times and Straits Times*
- Garnered Celent Model Bank 2021 award for Corporate Digital Bank
- Solutions include REST Architecture, IBM MQ Websphere, Tivoli Batch Scheduler Cron Job
- Developing API interfaces between different platforms that have been carefully designed to be relevant in the long run to accommodate agile user requirements
- Troubleshooting application bugs by analyzing Spring Boot logs, as well as designing and executing load testing scenarios and performing chaos testing to fulfill non-functional requirements (NFRs)
- Expertise in supervising project timeline, risks, budget, resources, vendor management, stakeholder communications and resolving conflict
- Independently led projects and managed a development team based in Xiamen, China

iOS Developer *Aug 2019 – Jan 2020*

- Worked on the development of a corporate banking application (IDEAL) using Swift 5 and Xcode IDE
- Champion of the SecureCodeWarrior competition on OWASP Top 10 vulnerabilities

Integrated Circuit Design Automation Intern *Singapore*
MediaTek *Jan 2018 – Jun 2018*

- Designed and implemented a Power Analytics Software for Integrated Circuit Design. The core engine utilized a custom clock tree graph database integrated with Synopsys software to generate fine-grained reports that identify power hungry zones
- Developed the backend using TCL and PERL to interface with Synopsys while the frontend was created with HTML, CSS, JavaScript, and jQuery for user interface
- Published an industry technical paper as the first author at SNUG 2018 in Singapore, titled "Power Doctor: Power Analytics Tool," in collaboration with MediaTek

Skillsets

Languages: Python, Matlab, R, Prolog, Swift, Perl, Tcl, HTML, CSS, JavaScript, Java, Verilog and ASM

Technologies: VSCode, Github, JIRA, Confluence, BitBucket, SourceTree, Jenkins, Linux (Red Hat Enterprise and Ubuntu), PyCharm, JupyterLab, Anaconda, Cloud Computing and LaTeX

Deep Learning Libraries: Pytorch, Pytorch-Lightning, Huggingface, Wandb and TensorBoardX

External Certifications

Industry Certificates: AWS Certified Solutions Architect – Associate, AWS Certified Cloud Practitioner, Scrum Alliance Certified ScrumMaster, NUS-ISS Cyber Security for ICT Professionals, IBM MQ Developer Essentials, Google Analytics Individual Qualification, Optiver Algorithmic Trading Workshop, DBS Advanced SRE Practitioner Certification

Online Course Certificates: Deep Learning Specialization, Generative AI with Large Language Models, Explainable AI (XAI) Specialization, Python for Everybody Specialization and Python 3 Programming Specialization